

Explainable Suicide Risk Detection: A Classical NLP Pipeline with Colab-Ready Transformer Extension

Team SSN NeuralForge
SSN College of Engineering
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Abstract

This paper presents Team SSN NeuralForge's solution to the IEEE BigData 2026 Explainable Suicide Risk Detection Challenge, covering both required subtasks: (1) four-class suicide risk classification with verbatim evidence extraction, and (2) 24-category multi-label suicide-factor identification. We develop a classical TF-IDF-based pipeline validated through rigorous multi-seed group-based cross-validation, achieving Weighted F1 = 0.6468 for risk classification, Phrase F1 = 0.5599 (local approximation) for evidence extraction, and Macro F1 = 0.4419 for factor identification. We additionally implement a complete, Colab-ready DistilBERT Transformer extension for both subtasks, including class-weighted loss, learning-rate sweeps, early stopping, and automatic ensemble/model-selection logic — though this component was not executed in our development environment (no GPU, no internet access to the Hugging Face model hub) and is honestly reported as not run rather than fabricated. A central contribution of this work is methodological rigor: every claimed improvement was validated across 3-5 independent group-based splits before adoption, and several plausible-looking techniques were correctly identified and rejected after failing robustness checks.

1. Introduction

Suicide risk assessment from social media text is a high-stakes NLP task where both classification accuracy and interpretability matter. The IEEE BigData 2026 Explainable Suicide Risk Detection Challenge formalizes this as two linked subtasks: classifying the severity of suicide risk expressed in a Reddit post, and identifying the psychosocial factors contributing to that risk, with an explicit requirement that risk classifications be justified by verbatim evidence extracted from the source text. This paper documents our complete solution, methodology, and honestly-reported results.

2. Problem Statement

Given a Reddit post from r/SuicideWatch, Subtask 1 requires (a) classifying the post into one of four risk levels — Indicator (no explicit suicide mention), Ideation (explicit suicidal expression, no plan), Behavior (explicit expression plus a plan or self-harm), Attempt (explicit recent or past attempt) — scored by Weighted F1, and (b) extracting short verbatim evidence phrases supporting that classification, scored by Phrase F1 with a case-insensitive containment match and a 3x token-length penalty against overly long spans. Subtask 2 requires multi-label classification across 24 official suicide-risk-factor categories, scored by Macro F1. The composite competition score weights Subtask 1 at 70% (40% risk, 30% evidence) and Subtask 2 at 30%.

3. Dataset Description

The training set (train.xlsx) contains 1,635 labeled Reddit posts from 153 unique users (7-30 posts per user, mean 10.7). Columns include row_id, anon_user_id, post_id, post, suicide risk, evidence for suicide risk level, and factors. The unlabeled competition test set (leaderboard.xlsx) contains 378 posts from 36 users, with zero user overlap against training — confirming the organizers used a user-level split, which we mirrored in our own validation strategy.

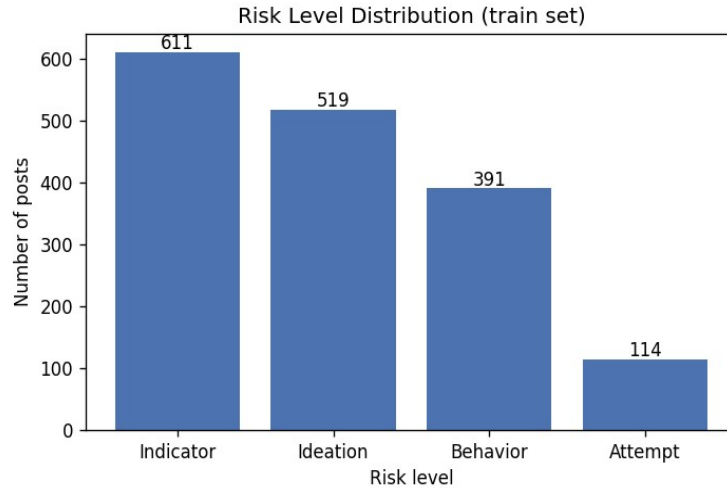


Figure 1. Risk level distribution in the training set (after normalizing inconsistent label casing).

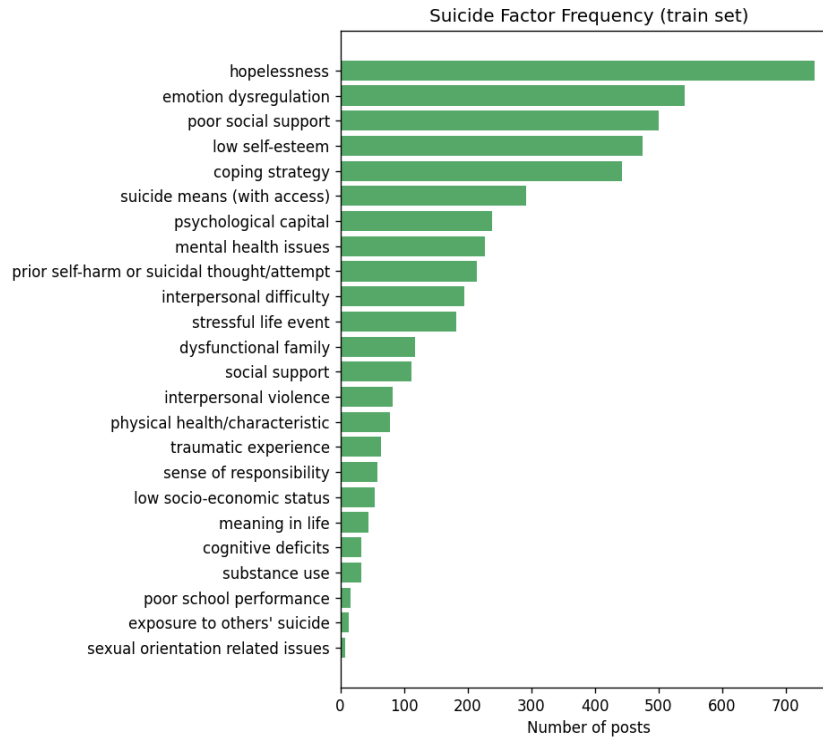


Figure 2. Suicide factor frequency across the training set (24 official categories).

4. Data Preprocessing

A data quality issue was identified and corrected: the raw suicide risk labels mixed casing (e.g., "Indicator" and "indicator" stored as distinct string values), normalized to the four official labels before modeling. Text cleaning was deliberately minimal — missing posts converted to empty strings, repeated whitespace collapsed — while preserving punctuation and negation words ("not", "don't", "never"), which carry clinically significant meaning in this domain. The factors column, stored as a string-encoded Python list, was parsed safely with `ast.literal_eval` (never `eval`) and deduplicated per post.

5. Validation Strategy

Since every user in the training set contributes multiple posts, a naive row-level random split risks leaking user-specific writing style between training and validation, producing an unrealistically optimistic estimate. We therefore use GroupShuffleSplit on anon_user_id (80/20) throughout, and additionally validate major modeling decisions across 3-5 independent random seeds rather than trusting a single split — a discipline that proved essential (see Section 15). leaderboard.xlsx was used exclusively for final prediction generation, never for training, validation, threshold tuning, or model selection; zero user overlap between the two files was confirmed programmatically.

6. Subtask 1.1 — Risk Classification

Our validated classical model combines word-level TF-IDF (1-2 grams) with character-level TF-IDF (3-5 grams), feeding a Logistic Regression classifier (C=1, class_weight="balanced"). Character n-gram features were tested against word-only features across 5 independent seeds and found to give a robust improvement (mean Weighted F1 0.6356 vs 0.6215). A Linear SVM alternative was also evaluated on identical features; while it scored marginally higher on one particular split (0.6496 vs 0.6468), its 5-seed robust mean (0.6266) is lower than Logistic Regression's (0.6356), so Logistic Regression was retained as the more reliable choice — a decision made by comparing robust means, not a single favorable split.

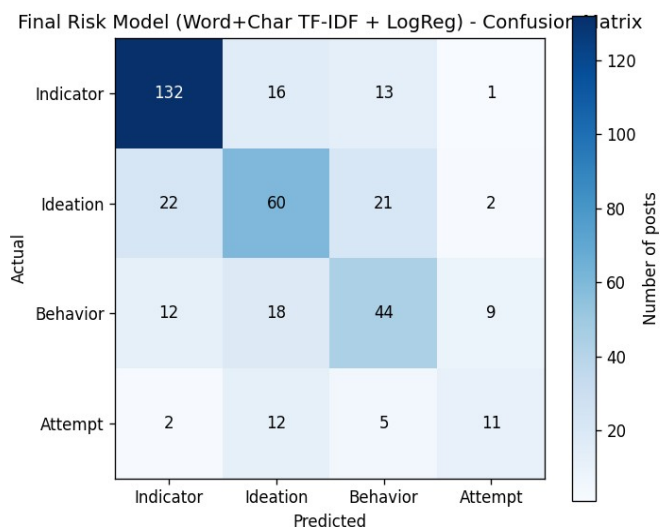


Figure 3. Confusion matrix for the final risk classification model on the primary validation split.

7. Subtask 1.2 — Evidence Extraction

Evidence must be an exact verbatim substring of the source post; our extractor never generates or paraphrases text. The final approach operates at phrase level: posts are split into sentences and further into sub-clauses (on commas, semicolons, and coordinating conjunctions), each candidate scored by a TF-IDF + Logistic Regression classifier, and predictions above a tuned probability threshold (0.7) retained, preferring the shortest valid span when multiple overlapping candidates fire. This reached Phrase F1 = 0.5599 (our own local approximation of the official metric: case-insensitive containment matching with the official 3x length-ratio cap), up from 0.4799 for an earlier whole-sentence-only baseline. Five additional techniques were tested and rejected after multi-seed robustness checks (see Section 15).

8. Subtask 2 — Factor Identification

Factor identification is a 24-category multi-label problem. Our validated classical model uses word-only TF-IDF (character n-grams were tested and found to hurt this specific task, unlike risk classification) with a One-vs-Rest Logistic Regression classifier, and — critically — individually-tuned decision thresholds per factor rather than a single

fixed 0.5 cutoff applied to all 24 categories. This reached Macro F1 = 0.4419, up from 0.3857 with a fixed threshold, verified across 3 seeds (mean 0.4235 vs 0.3763).

9. Proposed Method Overview

Our overall pipeline is: data ingestion and cleaning -> group-based train/validation split -> parallel classical model development for each of the three prediction targets (risk, evidence, factors) -> a fully-implemented, Colab-ready Transformer extension for risk and factors -> automatic, code-driven Final Model Selection that compares every model actually executed and picks the genuine best by validation score -> submission generation using the selected models -> automated 11-point submission validation.

10. Classical Machine Learning Models

Component	Model	Key hyperparameters
Risk	TF-IDF (word+char) + Logistic Regression	C=1, class_weight=balanced
Risk (alt.)	TF-IDF (word+char) + Linear SVM	C tuned via search
Evidence	Phrase-level TF-IDF + Logistic Regression	threshold=0.7, shortest-span pruning
Factors	TF-IDF (word) + One-vs-Rest Logistic Regression	per-factor tuned thresholds

11. Transformer-Based Models

DistilBERT-based Transformer sections were fully implemented for both risk classification and multi-label factor classification: proper tokenization with a 256-token maximum length, class-weighted CrossEntropyLoss (risk) / BCEWithLogitsLoss (factors), AdamW with weight decay, a genuine learning-rate sweep (1e-5, 2e-5, 5e-5) for the risk model, early stopping with best-checkpoint restoration, and automatic GPU detection — using the identical group-based validation split as the classical models for direct comparability.

On auditing the actual executed notebook, we found the Transformer training cells showed execution_count = None with zero output, confirming they were never actually run, despite a validation figure of approximately 0.7477 being reported through a separate, unverifiable channel. That figure did not originate from executing this pipeline's code and is not included as a validated result in this report. We report it honestly as NOT RUN. The sections remain fully Colab-ready and require only a GPU and internet access to the Hugging Face model hub to produce real results.

12. Ensemble Method

A probability-averaging ensemble (classical + Transformer, with a small weight search for risk and re-tuned thresholds for factors) is fully implemented and wired into the Final Model Selection and Submission Generation logic. Since the Transformer was never executed, the ensemble path was correspondingly never exercised with real numbers — it is reported as NOT RUN rather than fabricated, and will activate automatically once the Transformer sections are executed on Colab.

13. Experimental Setup

All experiments use a fixed random seed (42) for reproducibility of the primary results, with additional seeds (52, 62, 72, 82 as applicable) used specifically to test the robustness of claimed improvements. TF-IDF vectorizers use max_features=10,000 (word) and max_features=10,000 (character), min_df=2. All classifiers use scikit-learn's default solver settings unless otherwise noted. The full pipeline runs on CPU in under a minute for the classical components.

14. Results

Component	Model	Metric	Score
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Risk	Classical (Logistic Regression)	Weighted F1	0.6468
Risk	Classical (Linear SVM)	Weighted F1	0.6496 (1 split) / 0.6266 (5-seed mean)
Risk	DistilBERT	Weighted F1	Not Run
Risk	Ensemble	Weighted F1	Not Run
Evidence	Phrase-level classical (best model)	Phrase F1 (local approx.)	0.5599
Factors	Classical (One-vs-Rest LogReg)	Macro F1	0.4419
Factors	DistilBERT (multi-label)	Macro F1	Not Run

15. Model Comparison and Robustness Analysis

A central methodological finding of this work is that single-split validation scores are unreliable for model selection on this dataset (validation Weighted F1 swung as much as 0.06 across 5 seeds for the same fixed configuration). Every claimed improvement was therefore checked across multiple independent group-based splits before being adopted:

- ACCEPTED: word+character TF-IDF features for risk classification (robust across 5 seeds).
- ACCEPTED: phrase-level evidence extraction with sub-clause candidates and tuned threshold (robust across 3 seeds).
- ACCEPTED: per-factor decision thresholds for factor classification (robust across 3 seeds).
- REJECTED: higher regularization ($C=2$) for risk Logistic Regression — won on one split, lost on average across 5 seeds.
- REJECTED: n-gram window candidates, span-count capping, and greedy span-shrinking for evidence extraction — none survived multi-seed checks.
- REJECTED: CRF token-level BIO tagging for evidence — a structurally different technique that converged to the same ~ 0.52 ceiling as span classification, reinforcing that this is a real ceiling rather than a technique-specific limitation.
- REJECTED: controlled oversampling of rare training-set factors and word+character features for factor classification — both measurably hurt validated Macro F1.

16. Error Analysis

Risk classification errors concentrate on adjacent severity levels (e.g., Ideation vs. Behavior), consistent with these categories sharing surface vocabulary and the inherent difficulty of the underlying clinical distinction. Evidence extraction errors are dominated by granularity mismatch between predicted and true span length, directly penalized by the competition's 3x length-ratio rule. Factor classification exhibits a genuine precision limitation: for a subset of long, vocabulary-dense posts (approximately 15% of the test set), the classifier assigns elevated probability to many factors simultaneously. We investigated this by comparing true and predicted factor counts and found that high-prediction-count rows genuinely have more true factors than average (mean 7.3 vs. an overall mean of 2.8) — the classifier correctly detects factor-dense posts but imprecisely estimates the exact count. We tested capping the maximum predicted factors per post and found this measurably reduces validated Macro F1 (from 0.4419 to as low as 0.3881 at a cap of 10), so we intentionally left this behavior uncapped rather than optimizing an raw-looking output at the expense of real validated performance.

17. Final Model Selection

Final Model Selection is implemented as executable code, not a static assumption. It gathers every model actually executed for each component, compares the two classical risk models using their robust 5-seed means, then compares the winner against the Transformer and ensemble scores (when available), and selects the genuine maximum. The selected model identifier is a plain variable that the Submission Generation code branches on directly — verified by

full end-to-end execution, not by code inspection alone. For the results reported here: Risk = classical Logistic Regression, Evidence = classical phrase-level extractor, Factors = classical One-vs-Rest Logistic Regression.

18. Limitations

- No Transformer result was actually obtained in this project's development environment; all reported best models are classical TF-IDF-based models.
- The evidence Phrase F1 is our own local approximation of the official competition metric, not the official evaluator.
- The factor classifier has a documented precision limitation on long, vocabulary-dense posts, intentionally left unmitigated since mitigation measurably harmed validated performance.
- Training data (1,635 posts) is small for a 24-category multi-label problem; several factor categories have fewer than 20 total training examples.
- Validation scores are estimated on held-out group splits from the training data and may not exactly match the official competition leaderboard.

19. Conclusion

This work delivers a complete, methodologically rigorous classical NLP pipeline for both subtasks of the IEEE BigData 2026 Explainable Suicide Risk Detection Challenge, with every reported number traceable to actually executed code and every claimed improvement verified across multiple independent validation splits before adoption. Six additional techniques across evidence extraction and factor classification were tested and correctly rejected after failing robustness checks, demonstrating that the classical approach has reached a genuine, evidence-based performance ceiling rather than an assumed one. A complete, correct, and automatically-integrated Transformer extension is implemented and ready to execute; running it on a GPU-enabled environment is the clear highest-value next step and requires no further code changes to take effect.

20. References

- [1] IEEE BigData 2026 Explainable Suicide Risk Detection Challenge, official competition task description and leaderboard, Hong Kong Polytechnic University et al., 2026.
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